# Shadows at Noon

## 01 How can we save a shadow's length?



Mark the tip, then measure from the stick to the mark.

**Look closer.** Little puts a chalk mark at the long shadow's tip.

**A little explanation.** A shadow forms where an object blocks sunlight. Mark its tip, then measure from the standing stick to the mark. Use the same measuring stick and unit to compare both lengths.

**Try this.** With Big, measure from the stick's base to its shadow tip. Keep your eyes on the ground; never look at the sun.

## 02 Why is the noon shadow so short?



Around noon, the sun is near its highest point for the day.

**Look closer.** The same standing stick now has a short, crisp shadow.

**A little explanation.** Around noon, the sun is near its highest point in the sky for the day. Light reaches the ground from higher above, so the stick's shadow stays closer to its base. A lower morning or afternoon sun makes a longer shadow.

**Try this.** Compare the short shadow with the earlier chalk mark.

## 03 Which way does a shadow point?



The bright side faces the sun. The shadow points away from the light.

**Look closer.** One side of the stick is bright. The shadow is on the other side.

**A little explanation.** The bright side faces the sun. The stick blocks some sunlight, leaving a shadow that points away from the light. Follow the shadow along the ground.

**Try this.** Point from the bright side, past the stick, toward the shadow.

## 04 What clue can a short shadow give us?



A short noon shadow is a clue that the sun is high.

**Look closer.** Little keeps a long line and a short line in the notebook.

**A little explanation.** A short noon shadow is a clue that the sun is high. Compare the same standing stick in the same place, using the same unit. Chalk marks and notes help us remember what changed. Measuring can lead to another question.

**Try this.** Ask one new question about tomorrow's shadow.